INCH-POUND

MIL-L-44076A
30 July 1990
SUPERSEDING
MIL-L-44076
5 November 1981

MILITARY SPECIFICATION

LOWERING LINE ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of lowering line, with or without an adapter web.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-26 - Bag, Plastic, (General purpose)
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-636 - Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1670

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:44Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

11-1-2530 – Lowering Line Assembly

(Copies of drawings are available from the U.S. Army Natick Research, Development and Engineering Center, ATTN: STRNC-EMSS, Natick, MA 01760-5014.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of docuents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951-Standard practice for Commercial Packing

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:44Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Materials. Materials and hardware requirements for the lowering line assembly shall be in accordance with Drawing 11-1-2530, subsidiary drawings, and as specified herein. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3 Construction and assembly. The end item shall conform to the requirements of this specification and to the designs, details, and dimensions shown or specified on the applicable drawings.

3.3.1 Stitch patterns. Automatic machine stitch patterns providing strength equal to the patterns specified on Drawing 11-1-2530 which conform to the overall dimensions, and which do not adversely affect form, fit, and function of the end item may be used in lieu of those specified, upon approval of the contracting officer. The contractor may be required to provide proof of equivalency.

3.4 Workmanship. The end item shall conform to the quantity of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:44Z
Check the source to verify that this is the current version before use.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the ccontractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The insection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2 and 4,4.3. Any nonconformance shall be cause for the rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4,4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of lowering line assemblies. The sample unit shall one lowering line assembly. The inspection level shall shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units shall be 1.5 for major defects and 6.5 for total (major and minor combined) defects.

Table I. End item visual defects

Examine	Defect	Classification Major	Minor
Material	Wrong color		201
Webbing	Not specified type or class	101	
	Cut, tear, hole, abrasion, or weak area	102	
	Edges not seared as specified		202
Metal components	Fractured, malformed, or bent	103	
	Burrs or sharp edges	104	
	Protective finish omitted or not as specified, any corroded area	105	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:44Z
Check the source to verify that this is the current version before use.

Table I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Thread	Not type, class, or size specified	106	
	Not specified color		203
Stitches and stitching	Stitching pattern not as specified	107	
	Any row of stitching missing	108	
	One stitch per inch under or over minimum and maximum specified		204
	Two or more stitches per inch under or over minimum and maximum specified	109	
	Thread breaks, bobbin runouts, or two or more consecutive skipped or runoff stitches, not overstitched or restitched as specified		205
	Ends of stitching not backstitched or secured as specified		206
	Tension loose resulting in loose top or bottom thread, or tight tension causing puckering of material at stitch line	110	
	Needle chews or bunched stitching	111	
Construction and assembly	Any component missing, or not assembled or joined as specified	112	
Marking	Omitted; incorrect; illegible; wrong location, size, or color		207
Workmanship	Grease, oil, ink smears, dirt, or other foreign matter; thread ends not trimmed throughout		208

4.4.3 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on Drawing 11-1-2530. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of lowering line assemblies. The sample unit shall be on lowering line assembly. The inspection level shall be S-3 and the AQL expressed in terms of defects per hundred units, shall be 4.0.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:44Z
Check the source to verify that this is the current version before use.

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation. Each lowering line assembly or lowering line assembly with adapter web attached shall be compactly arranged and then unit packed in a close-fitting 0.002-inch thick polyethylene film bag conforming to type I or II, style 1 of PPP-B-26. Prior to or during the final heat seal closure operation, excess air within the bag shall be expelled.

5.1.2 Commercial preservation. Each lowering line assembly or lowering line assembly with adapter web attached shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. Eighty lowering line assemblies or 60 lowering line assemblies with adapter webs attached, preserved as specified in 5.1 shall be packed in a snug-fitting shipping container conforming to overseas type, style A or 3, grade A or B, type 2 load of PPP-B-601. Each shipping container shall be closed and reinforced in accordance with the appendix of PPP-B-601.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:44Z
Check the source to verify that this is the current version before use.

5.2.2 Level B packing. Eighty lowering line assemblies or 60 lowering line assemblies with adapter webs attached, preserved as specified in 5.1 shall be packed in a snug-fitting shipping container conforming to style RSC-L, grade V3c, V3s, or V4s of PPP-B-636. The inside of each fiberboard container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety SW or DW, grade V3c or V15c of PPP-B-636. Each shipping container shall be closed in accordance with method III and waterproofed in accordance with method V as specified in the appendix of PPP-3-636.

5.2.3 Commercial packing. A quantity of lowering line assemblies or lowering line assemblies with adapter webs attached, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Lowering line assemblies are used as the standard method of lowering all individual parachutist's equipment loads when jumping with personnel parachutes.

6.2 Acquisition requirements. Acquisition documents must specify the following:

Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents, referenced (see 2.1.1 and 2.2).
When a first article is required (see 3.1, 4.3, and 6.3).
d. Levels of preservation and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

Equipment loads
Parachutists

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:44Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-44076A	1990 July 30

3. DOCUMENT TITLE
LOWERING LINE ASSEMBLY

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
U.S. Army Natick RD&E Center	508-651-5221	256-5221
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-ES Natick, MA 01760-5014	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 NATICK SUP, APR 90 Previous editions are obsolete. 198/290

Source: https://assist.dla.mil - Downloaded: 2026-10-21T23:44Z
Check the source to verify that this is the current version before use.